

DIGITAL ELECTRONICS

Unit-2: Logic Gates and Boolean Algebra

Comprehensive Study Notes

HPU BCA Semester-I

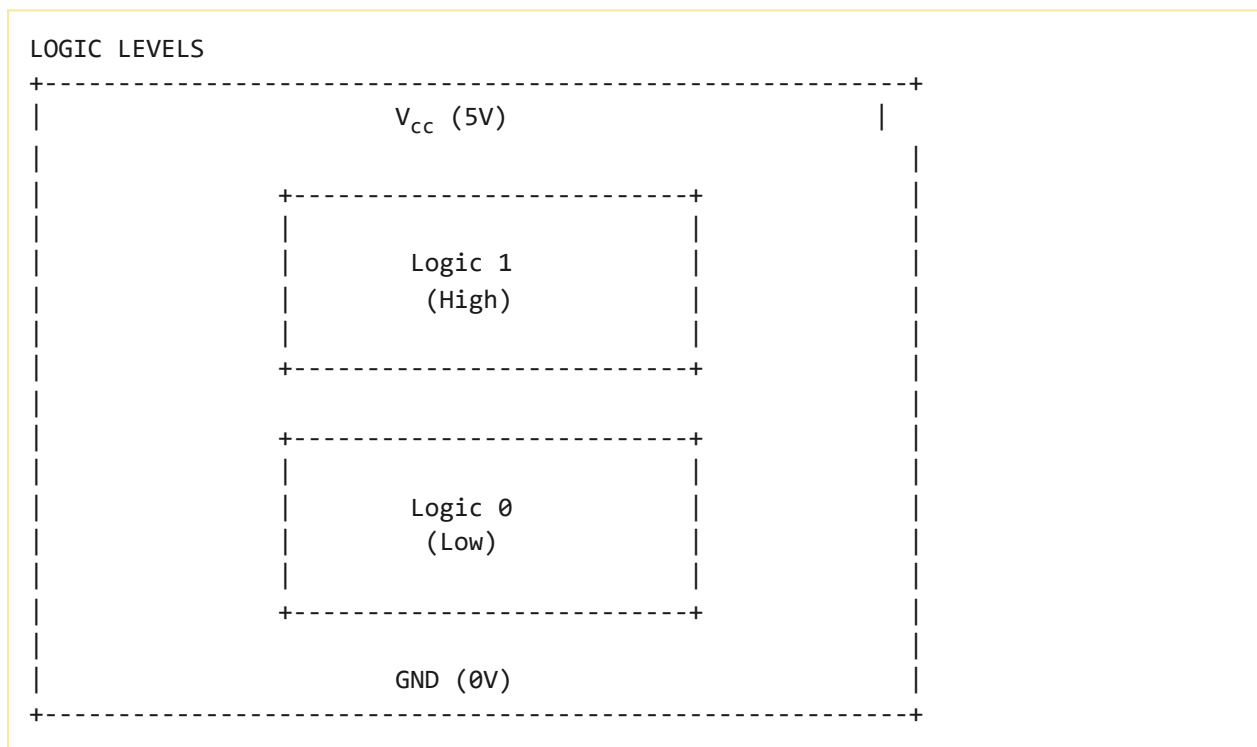
Topic 1: Introduction to Logic Gates

1.1 What are Logic Gates?

A **logic gate** is a basic building block of digital circuits that performs a logical operation on one or more binary inputs and produces a single binary output. Logic gates are the fundamental components of all digital systems, including computers, calculators, and other electronic devices.

Logic Levels

- **Logic 0 (Low):** Typically represents 0V
- **Logic 1 (High):** Typically represents 5V (or V_{CC})



Topic 2: Basic Logic Gates

2.1 NOT Gate (Inverter)

Definition: A NOT gate has **one input** and **one output**. The output is the **inverse** (complement) of the input.

Symbol: Input -----> [NOT] -----> Output

Logic Diagram:

```

+---+
|   |
A--+NOT+--- Y
|   |
+---+
  
```

Input (A)

Output (Y)

0	1
1	0

Boolean Expression:

$Y = A'$ (or $Y = \text{NOT } A$)

Applications: Complementing signals, creating oscillators, inverting logic levels.

2.2 AND Gate

Definition: An AND gate has **two or more inputs** and **one output**. The output is **High (1)** only when **all** inputs are High (1).

Symbol: Input A ----+
 +---[AND]----> Output
 Input B ----+

Logic Diagram:

```

+-----+
A--+   |
   |   | AND +---- Y
B--+   |
+-----+
```

Input A	Input B	Output (Y)
0	0	0
0	1	0
1	0	0
1	1	1

Boolean Expression:

$$Y = A \cdot B \quad (\text{or } Y = A \text{ AND } B)$$

Applications: Enable/disable circuits, data selection, logic control.

2.3 OR Gate

Definition: An OR gate has **two or more inputs** and **one output**. The output is **High (1)** when **any** input is High (1).

Symbol: Input A ----+
 +---[OR]---> Output
 Input B ----+

Logic Diagram:

```

+-----+
A--+   |
|  OR  +----- Y
B--+   |
+-----+
  
```

Input A	Input B	Output (Y)
0	0	0
0	1	1
1	0	1
1	1	1

Boolean Expression:

$$Y = A + B \quad (\text{or } Y = A \text{ OR } B)$$

Applications: Combining multiple signals, priority circuits, alarm systems.

Topic 3: Universal Logic Gates

3.1 NAND Gate (NOT AND)

Definition: A NAND gate is an **AND gate followed by a NOT gate**. The output is the **inverse** of the AND operation.

Symbol: Input A ----+
 +--[NAND]----> Output
 +----+
 Input B ----+

Logic Diagram:

```

+-----+
A--+   |
  |    |
  | NAND +---- Y
B--+   |
+-----+
  
```

Alternative: AND + NOT

```

+---+ +---+
A--+ | | |
  |  |AND+---+NOT+--- Y
B--+ | | |
+---+ +---+
  
```

Input A	Input B	Output (Y)
0	0	1
0	1	1
1	0	1
1	1	0

Boolean Expression:

$$Y = (A \cdot B)' \quad (\text{or } Y = A \text{ NAND } B)$$

Why NAND is Universal: NAND gates can be used to create **ANY** other logic gate. This is why NAND-based designs are common in ICs.

3.2 NOR Gate (NOT OR)

Definition: A NOR gate is an **OR gate followed by a NOT gate**. The output is the **inverse** of the OR operation.

Symbol: Input A ----+
 +---[NOR]----> Output
 Input B ----+

Logic Diagram:

```

+-----+
A--+   |
  | NOR +---- Y
B--+   |
+-----+
```

Alternative: OR + NOT

```

+---+ +---+
A--+ |   |   |
  |OR +---+NOT+--- Y
B--+ |   |   |
+---+ +---+
```

Input A	Input B	Output (Y)
0	0	1
0	1	0
1	0	0
1	1	0

Boolean Expression:

$$Y = (A + B)'$$

(or $Y = A \text{ NOR } B$)

Topic 4: XOR and XNOR Gates

4.1 XOR Gate (Exclusive OR)

Definition: An XOR gate outputs **High (1)** when the **inputs are different** (one 0 and one 1).

Symbol: Input A ----+
 +---[XOR]----> Output
 Input B ----+

Logic Diagram:

```

+-----+
A--+   |
| XOR  +----- Y
B--+   |
+-----+
```

Input A	Input B	Output (Y)
0	0	0
0	1	1
1	0	1
1	1	0

Boolean Expression:

$$Y = A \oplus B = A'B + AB'$$

Applications: Comparator circuits (checking equality), parity generators and checkers, binary addition (Half Adder), controlled inverters.

4.2 XNOR Gate (Exclusive NOR)

Definition: An XNOR gate outputs **High (1)** when the **inputs are same** (both 0 or both 1). It is the inverse of XOR.

Symbol: Input A ----+
 +---[XNOR]----> Output
 Input B ----+

Logic Diagram:

```

+-----+
A--+      |
|   XNOR  +---- Y
B--+      |
+-----+
```

Input A	Input B	Output (Y)
0	0	1
0	1	0
1	0	0
1	1	1

Boolean Expression:

$$Y = (A \oplus B)' = AB + A'B'$$

Applications: Comparator circuits (checking equality), parity checkers, equality detection, controlled inverters.

Topic 5: Gate Summary Table

Gate	Symbol	Boolean Expression	Truth Table (A B -> Output)
NOT	[NOT]	$Y = A'$	0->1, 1->0
AND	[AND]	$Y = A \cdot B$	00->0, 01->0, 10->0, 11->1
OR	[OR]	$Y = A + B$	00->0, 01->1, 10->1, 11->1
NAND	[NAND]	$Y = (A \cdot B)'$	00->1, 01->1, 10->1, 11->0
NOR	[NOR]	$Y = (A + B)'$	00->1, 01->0, 10->0, 11->0
XOR	[XOR]	$Y = A \oplus B$	00->0, 01->1, 10->1, 11->0
XNOR	[XNOR]	$Y = (A \oplus B)'$	00->1, 01->0, 10->0, 11->1

Topic 6: Boolean Algebra

6.1 Introduction

Boolean Algebra is a mathematical system that deals with **binary variables** and **logical operations**. It is named after George Boole, who developed it in the 19th century.

Boolean algebra is the foundation of digital logic design.

Boolean Variables

- Represented by letters (A, B, C, X, Y, Z)
- Only two values: 0 (False) and 1 (True)

Basic Operations

- **NOT (Complement):** A' or $\neg A$
- **AND:** $A \cdot B$
- **OR:** $A + B$

6.2 Basic Boolean Laws

1. Identity Law

Law	Expression
OR Identity	$A + 0 = A$
AND Identity	$A \cdot 1 = A$

2. Null Law (Zero Law)

Law	Expression
OR Null	$A + 1 = 1$
AND Null	$A \cdot 0 = 0$

3. Idempotent Law

Law	Expression
OR Idempotent	$A + A = A$
AND Idempotent	$A \cdot A = A$

4. Involution Law (Double Complement)

Law	Expression
Double Complement	$(A')' = A$

5. Complement Law

Law	Expression

OR Complement	$A + A' = 1$
AND Complement	$A \cdot A' = 0$

6. Commutative Law

Law	Expression
OR Commutative	$A + B = B + A$
AND Commutative	$A \cdot B = B \cdot A$

7. Associative Law

Law	Expression
OR Associative	$(A + B) + C = A + (B + C)$
AND Associative	$(A \cdot B) \cdot C = A \cdot (B \cdot C)$

8. Distributive Law

Law	Expression
AND over OR	$A \cdot (B + C) = (A \cdot B) + (A \cdot C)$
OR over AND	$A + (B \cdot C) = (A + B) \cdot (A + C)$

9. Absorption Law

Law	Expression
Absorption 1	$A + (A \cdot B) = A$
Absorption 2	$A \cdot (A + B) = A$

6.3 Solved Examples: Boolean Laws

Example 1: Simplify $A + A \cdot B$

$$A + A \cdot B = A + B \quad (\text{Using Absorption Law})$$

Example 2: Simplify $A \cdot B + A \cdot B'$

$$\begin{aligned} A \cdot B + A \cdot B' &= A \cdot (B + B') && (\text{Distributive Law}) \\ &= A \cdot 1 && (\text{Complement Law}) \\ &= A && (\text{Identity Law}) \end{aligned}$$

Example 3: Simplify $(A + B) \cdot (A + B')$

$$\begin{aligned} (A + B) \cdot (A + B') &= A + (B \cdot B') && (\text{Distributive Law}) \\ &= A + 0 && (\text{Complement Law}) \\ &= A && (\text{Identity Law}) \end{aligned}$$

Topic 7: DeMorgan's Theorem

7.1 Statement

DeMorgan's Theorem provides a way to simplify Boolean expressions involving complementation of compound statements. It is one of the most important theorems in digital logic.

Theorem 1:

$$(A \cdot B)' = A' + B'$$

"The complement of an AND is the OR of complements."

Theorem 2:

$$(A + B)' = A' \cdot B'$$

"The complement of an OR is the AND of complements."

7.2 General Forms (for n variables)

Form	Expression
AND Form	$(A1 \cdot A2 \cdot \dots \cdot An)' = A1' + A2' + \dots + An'$
OR Form	$(A1 + A2 + \dots + An)' = A1' \cdot A2' \cdot \dots \cdot An'$

7.3 Proof of DeMorgan's Theorem

Proof of $(A \cdot B)' = A' + B'$

A	B	$A \cdot B$	$(A \cdot B)'$	A'	B'	$A' + B'$
0	0	0	1	1	1	1
0	1	0	1	1	0	1
1	0	0	1	0	1	1
1	1	1	0	0	0	0

All columns match → Theorem is proven!

Proof of $(A + B)' = A' \cdot B'$

A	B	$A + B$	$(A + B)'$	A'	B'	$A' \cdot B'$
0	0	0	1	1	1	1
0	1	1	0	1	0	0
1	0	1	0	0	1	0
1	1	1	0	0	0	0

All columns match → Theorem is proven!

7.4 Applications of DeMorgan's Theorem

Application	Description
Logic Simplification	Simplify complex Boolean expressions
Gate Conversion	Convert NAND to NOR and vice versa

Circuit Design	Design circuits using NAND/NOR gates only
Boolean Algebra	Prove logical equivalences

Topic 8: Boolean Functions and Their Truth Tables

8.1 Boolean Functions

A Boolean function is an **expression formed using binary variables, constants (0 and 1), and logical operations.**

Example: $F(A, B, C) = A \cdot B + B' \cdot C$

8.2 Creating Truth Tables

Steps:

1. List all possible combinations of input values
2. Calculate the output for each combination
3. Fill in the truth table

8.3 Solved Examples

Example 1: $F(A, B) = A \cdot B$

A	B	A·B (F)
0	0	0
0	1	0
1	0	0
1	1	1

Example 2: $F(A, B) = A + B'$

A	B	B'	A + B' (F)
0	0	1	1
0	1	0	0
1	0	1	1
1	1	0	1

Example 3: $F(A, B) = A'B' + AB'$ (XOR)

A	B	A'	B'	A'B'	AB'	F
0	0	1	1	0	0	0
0	1	1	0	0	1	1
1	0	0	1	1	0	1
1	1	0	0	0	0	0

Topic 9: SOP, POS Forms, Minterms & Maxterms

9.1 Minterms

A **minterm** is a product term (AND) that includes all variables in either complemented or uncomplemented form. For n variables, there are 2^n minterms.

Variables (A, B)	Minterm	Notation
0 0	$A'B'$	m_0
0 1	$A'B$	m_1
1 0	AB'	m_2
1 1	AB	m_3

9.2 Maxterms

A **maxterm** is a sum term (OR) that includes all variables in either complemented or uncomplemented form.

Variables (A, B)	Maxterm	Notation
0 0	$A + B$	M_0
0 1	$A + B'$	M_1
1 0	$A' + B$	M_2
1 1	$A' + B'$	M_3

9.3 SOP (Sum of Products) Form

SOP is a Boolean expression as a sum (OR) of product terms (AND). It is formed by ORing minterms where the output is 1.

Example: $F = A'B + AB'$ (SOP form)

9.4 POS (Product of Sums) Form

POS is a Boolean expression as a product (AND) of sum terms (OR). It is formed by ANDing maxterms where the output is 0.

Example: $F = (A+B) \cdot (A'+B')$ (POS form)

Questions

Short Answer Questions (2-5 Marks)

1. Define logic gate.
2. Draw the symbol and truth table for AND gate.
3. Draw the symbol and truth table for OR gate.
4. Draw the symbol and truth table for NOT gate.
5. What is a NAND gate? Write its truth table.
6. What is a NOR gate? Write its truth table.
7. State DeMorgan's theorems.
8. What is XOR gate? Give its truth table.
9. What is XNOR gate? Give its truth table.
10. Write the Boolean expression for XOR gate.

Long Answer Questions (10 Marks)

1. Explain all basic logic gates with symbols and truth tables.
2. Explain NAND and NOR gates. Why are they called universal gates?
3. Explain XOR and XNOR gates with truth tables and applications.
4. State and prove DeMorgan's theorem.
5. Simplify the following Boolean expression: $A'B + AB' + A'B'$
6. Simplify the following Boolean expression using Boolean algebra: $A \cdot (A + B)$

HPU Exam-Based Questions

1. Write the truth table for NAND gate.
2. State DeMorgan's theorem.
3. Define universal gates.

Objective Questions (1 Mark)

1. The output of AND gate is 1 when:
 - a. All inputs are 0

- b. All inputs are 1
- c. Any input is 1
- d. None

Answer: (b) All inputs are 1

2. The output of OR gate is 1 when:

- a. All inputs are 0
- b. All inputs are 1
- c. Any input is 1
- d. None

Answer: (c) Any input is 1

3. The output of NOT gate is:

- a. Same as input
- b. Complement of input
- c. Always 0
- d. Always 1

Answer: (b) Complement of input

4. NAND gate is AND gate followed by:

- a. AND gate
- b. OR gate
- c. NOT gate
- d. NOR gate

Answer: (c) NOT gate

5. NOR gate is OR gate followed by:

- a. AND gate
- b. OR gate
- c. NOT gate
- d. NAND gate

Answer: (c) NOT gate

6. XOR gate is also called:

- a. OR gate
- b. AND gate
- c. Exclusive OR gate
- d. Exclusive NOR gate

Answer: (c) Exclusive OR gate

7. DeMorgan's theorem states that $(A \cdot B)' =$

- a. $A' \cdot B'$
- b. $A' + B'$

c. $A' \cdot B'$

d. $A' \cdot B$

Answer: (b) $A' + B'$

Quick Revision

Logic Gates Summary

Gate Name	Boolean Expression	Truth Table
NOT	$Y = A'$	0- $\rightarrow$ 1, 1- $\rightarrow$ 0
AND	$Y = A \cdot B$	00- $\rightarrow$ 0, 11- $\rightarrow$ 1
OR	$Y = A + B$	00- $\rightarrow$ 0, 01- $\rightarrow$ 1
NAND	$Y = (A \cdot B)'$	00- $\rightarrow$ 1, 11- $\rightarrow$ 0
NOR	$Y = (A + B)'$	00- $\rightarrow$ 1, 01- $\rightarrow$ 0
XOR	$Y = A \oplus B$	00- $\rightarrow$ 0, 01- $\rightarrow$ 1
XNOR	$Y = (A \oplus B)'$	00- $\rightarrow$ 1, 01- $\rightarrow$ 0

Boolean Laws Summary

Law	Expression
Identity	$A + 0 = A, A \cdot 1 = A$
Null	$A + 1 = 1, A \cdot 0 = 0$
Idempotent	$A + A = A, A \cdot A = A$
Complement	$A + A' = 1, A \cdot A' = 0$
Involution	$(A')' = A$
Commutative	$A + B = B + A, A \cdot B = B \cdot A$
Associative	$(A + B) + C = A + (B + C)$
Distributive	$A \cdot (B + C) = A \cdot B + A \cdot C$

Absorption	$A + A \cdot B = A, A \cdot (A + B) = A$
DeMorgan	$(A \cdot B)' = A' + B', (A + B)' = A' \cdot B'$

Important Full Forms

Abbreviation	Full Form
AND	-
OR	-
NOT	-
NAND	NOT AND
NOR	NOT OR
XOR	Exclusive OR
XNOR	Exclusive NOR